IT3011 - Theory & Practices in Statistical Modelling

Population Variances

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$(1-\alpha)100\%$ Confidence Interval for variance

Case 01 - μ is known

Let $x_1, x_2, x_3, \ldots, x_n$ be a random sample of size n from $N(\mu, \sigma^2)$ .We know that for a sample of a population,

$$\sum_{i=1}^{n} Z_i^2 = \sum_{i=1}^{n} \left(\frac{X_i - \mu}{\sigma}\right)^2 \sim \chi^2_{(n)}$$

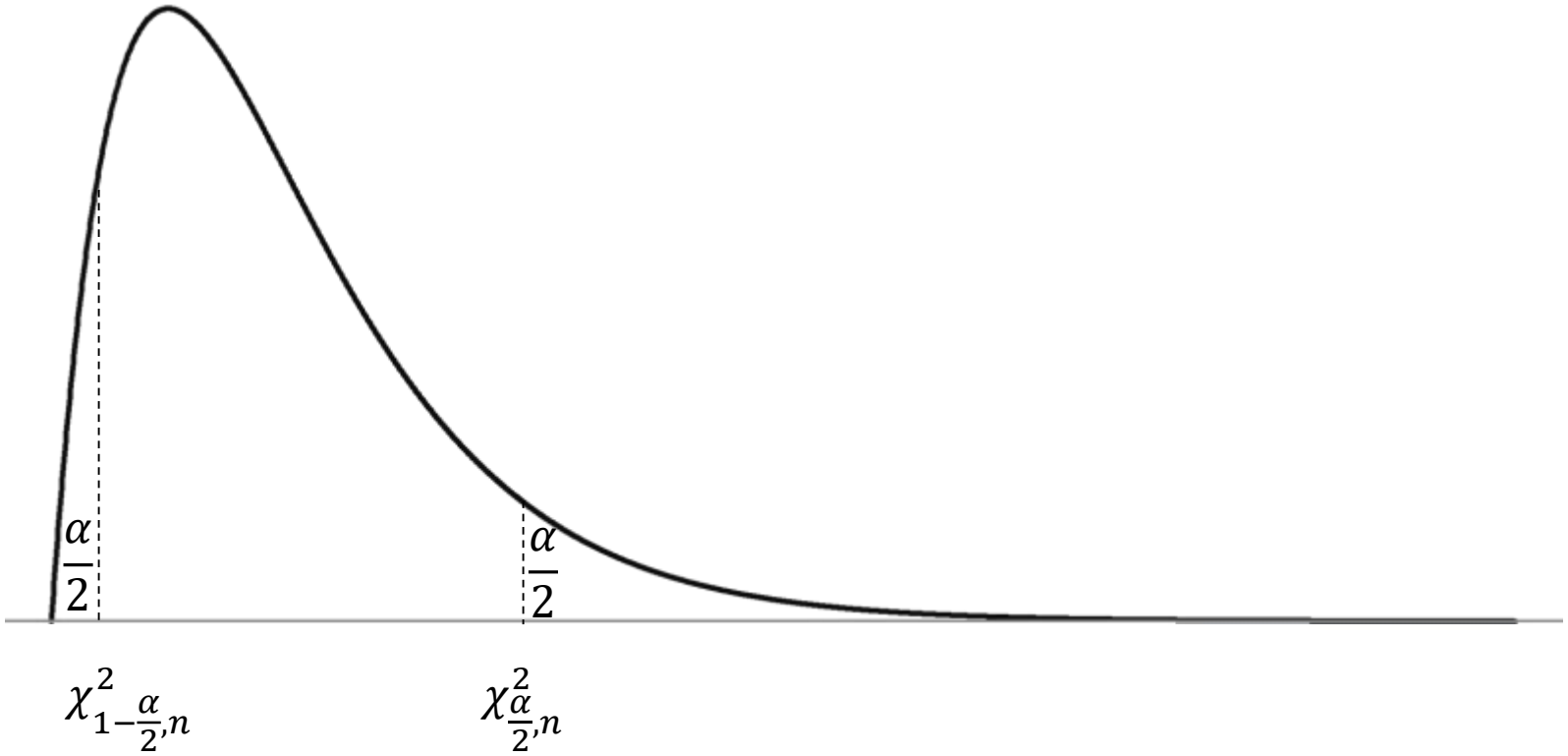
Consider this as the pivotal quantity. Then,

$$P\left(L \leq \sum_{i=1}^{n} \left(\frac{X_i - \mu}{\sigma}\right)^2 \leq U\right) = 1 - \alpha$$

$(1-\alpha)100\%$ Confidence Interval for variance

Case 01 - μ is known

In the Chi-Squared distribution,



Distribution goes from here to there

$(1-\alpha)100\%$ Confidence Interval for variance

Case 01 - μ is known

Then,

$$P\left(\chi^2_{1-\frac{\alpha}{2},n} \leq \sum_{i=1}^{n}\left(\frac{X_i-\mu}{\sigma}\right)^2 \leq \chi^2_{\frac{\alpha}{2},n}\right) = 1-\alpha$$

$$P\left(\frac{1}{\chi^2_{1-\frac{\alpha}{2},n}} \geq \frac{\sigma^2}{\sum_{i=1}^{n}(X_i-\mu)^2} \geq \frac{1}{\chi^2_{\frac{\alpha}{2},n}}\right) = 1-\alpha$$

$$P\left(\frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\chi^2_{1-\frac{\alpha}{2},n}} \geq \sigma^2 \geq \frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\chi^2_{\frac{\alpha}{2},n}}\right) = 1-\alpha$$

$$P\left(\frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\chi^2_{\frac{\alpha}{2},n}} \leq \sigma^2 \leq \frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\chi^2_{1-\frac{\alpha}{2},n}}\right) = 1-\alpha$$

$(1-\alpha)100\%$ Confidence Interval for variance

Case 01 - μ is known

Then $(1-\alpha)100\%$ confidence interval for variance when the population mean is known is,

$$\left[\frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\chi^2_{\frac{\alpha}{2},n}}, \frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\chi^2_{1-\frac{\alpha}{2},n}}\right]$$

$(1-\alpha)100\%$ Confidence Interval for variance

Case 02 - μ is unknown

We know that for a sample of a population,

$$\sum_{i=1}^{n}\left(\frac{X_i-\bar{X}}{\sigma}\right)^2=\frac{(n-1)S^2}{\sigma^2}\sim\chi^2_{(n-1)}$$

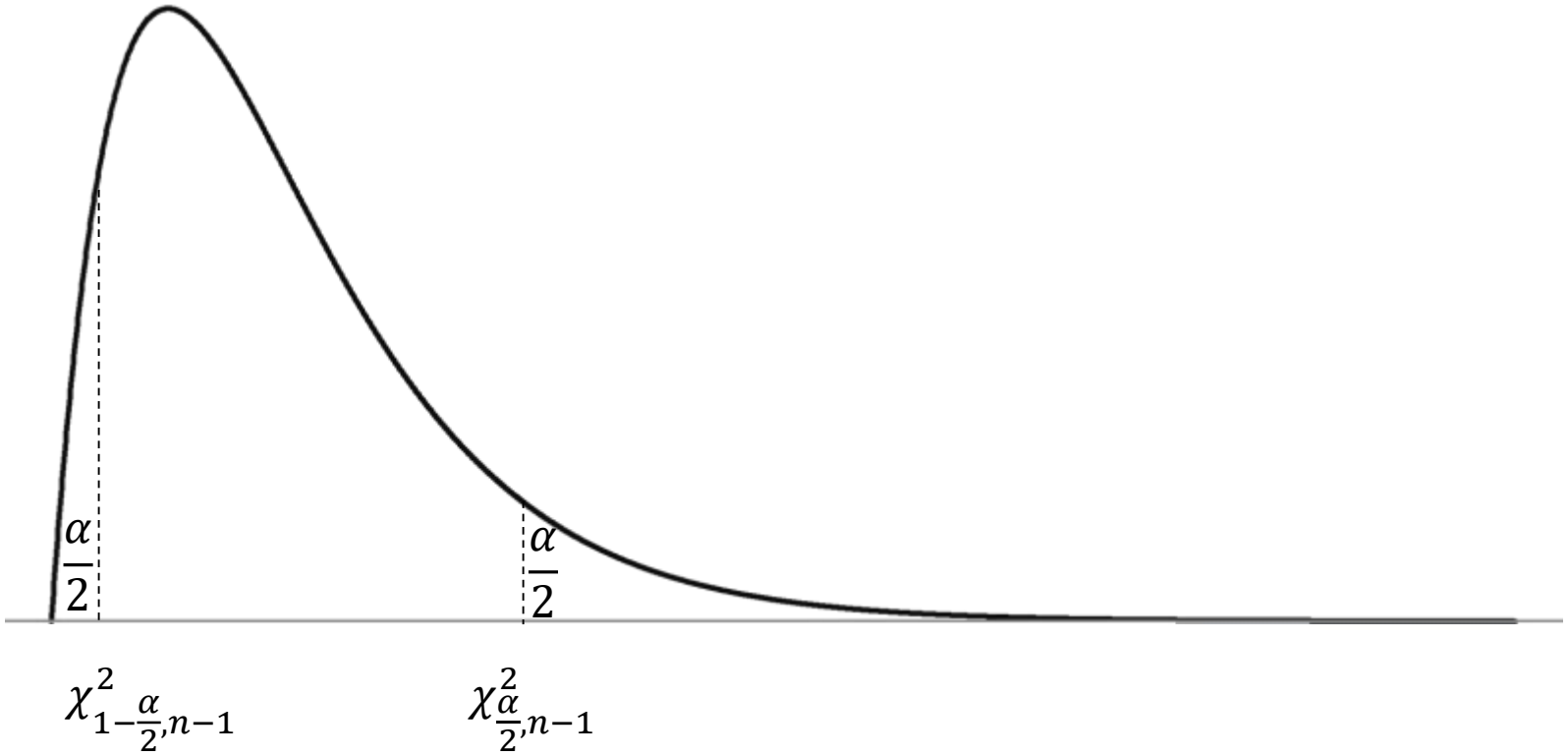
Consider this as the pivotal quantity. Then,

$$P\left(L\leq\frac{(n-1)S^2}{\sigma^2}\leq U\right)=1-\alpha$$

$(1-\alpha)100\%$ Confidence Interval for variance

Case 02 - μ is unknown

In the Chi-Squared distribution,



Distribution goes from here to there

$(1-\alpha)100\%$ Confidence Interval for variance

Case 02 - μ is unknown

Then,

$$P\left(\chi^2_{1-\frac{\alpha}{2},n-1} \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi^2_{\frac{\alpha}{2},n-1}\right) = 1-\alpha$$

$$P\left(\frac{1}{\chi^2_{1-\frac{\alpha}{2},n-1}} \geq \frac{\sigma^2}{(n-1)S^2} \geq \frac{1}{\chi^2_{\frac{\alpha}{2},n-1}}\right) = 1-\alpha$$

$$P\left(\frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2},n-1}} \geq \sigma^2 \geq \frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2},n-1}}\right) = 1-\alpha$$

$$P\left(\frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2},n-1}} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2},n-1}}\right) = 1-\alpha$$

$(1-\alpha)100\%$ Confidence Interval for variance

Case 02 - μ is unknown

Then $(1-\alpha)100\%$ confidence interval for variance when the population mean is unknown is,

$$\left[\frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2},n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2},n-1}}\right]$$

$(1-\alpha)100\%$ Confidence Interval for variance

Ex:-

The following are the weights, in decagrams, of 10 packages of grass seed distributed by a certain company:

$$46.4, 46.1, 45.8, 47, 46.1, 45.9, 45.8, 46.9, 45.2, 46$$

Find the 95% confidence interval for the variance of all such packages of grass seed distributed by this company, assuming a normal population.

$$S^2 = 0.2862$$

$$\chi^2_{0.025,9} = 19.0$$

$$\chi^2_{0.975,9} = 2.70$$

$$\left[\frac{9 \times 0.2862}{19}, \frac{9 \times 0.2862}{2.70}\right] = [0.135, 0.954]$$

Chi Squared Table

Degrees of Freedom	Area to the Right of Critical Value							
	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01
1	—	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.071	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217
13	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932
22	9.542	10.982	12.338	14.042	30.813	33.924	36.781	40.289
23	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638
24	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980
25	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314
26	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642
27	12.879	14.573	16.151	18.114	36.741	40.113	43.194	46.963
28	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278
29	14.257	16.047	17.708	19.768	39.087	42.557	45.722	49.588
30	14.954	16.791	18.493	20.599	40.256	43.773	46.979	50.892

$(1-\alpha)100\%$ Confidence Interval for ratio of two population variances

We know that for two samples of two populations where means are unknown,

$$\frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2} \sim F(n_1 - 1, n_2 - 1)$$

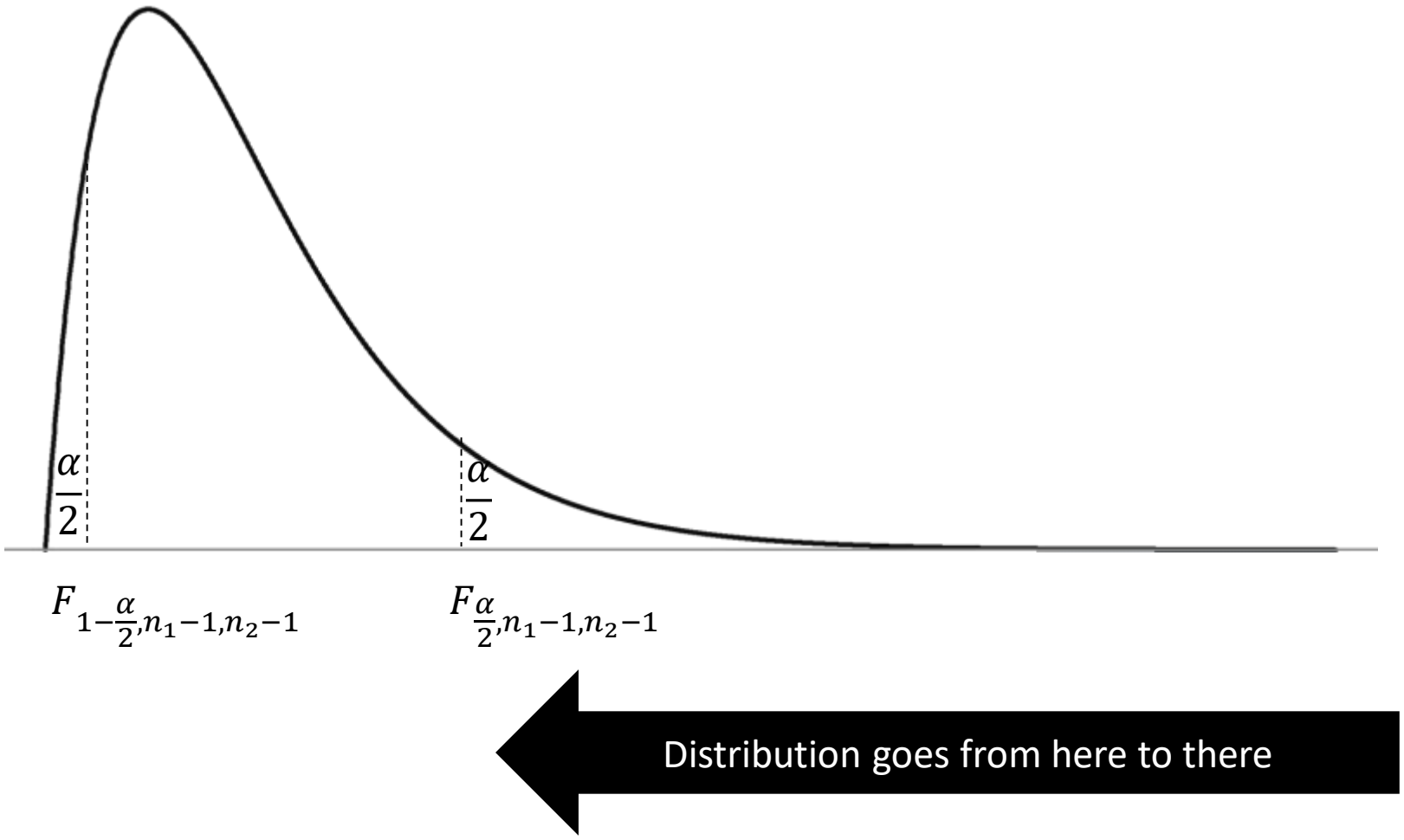
Where n_1 and n_2 are sample sizes and σ_1^2, S_1^2 are variances for the first population and its sample as well as σ_2^2, S_2^2 are variances for the second population and its sample.

Consider this as the pivotal quantity. Then,

$$P\left(L \leq \frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2} \leq U\right) = 1 - \alpha$$

$(1-\alpha)100\%$ Confidence Interval for ratio of two population variances

In the F distribution,



$(1-\alpha)100\%$ Confidence Interval for ratio of two population variances

Then,

$$P\left(F_{1-\frac{\alpha}{2},n_1-1,n_2-1} \leq \frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2} \leq F_{\frac{\alpha}{2},n_1-1,n_2-1}\right) = 1-\alpha$$

$$P\left(\frac{S_2^2}{S_1^2} F_{1-\frac{\alpha}{2},n_1-1,n_2-1} \leq \frac{\sigma_2^2}{\sigma_1^2} \leq \frac{S_2^2}{S_1^2} F_{\frac{\alpha}{2},n_1-1,n_2-1}\right) = 1-\alpha$$

Then $(1-\alpha)100\%$ confidence interval for ration of two population variances $\left(\frac{\sigma_2^2}{\sigma_1^2}\right)$ is,

$$\left[\frac{S_2^2}{S_1^2} F_{1-\frac{\alpha}{2},n_1-1,n_2-1}, \qquad \frac{S_2^2}{S_1^2} F_{\frac{\alpha}{2},n_1-1,n_2-1}\right]$$

$(1-\alpha)100\%$ Confidence Interval for ratio of two population variances

Ex:-

Two independent random samples of water bottles were drawn from 2 automatic bottle filling machines in a factory. Volume of water in each selected bottle was measured. 25 bottles from machine 1 had sample variance S_1^2 =1.63 and 16 observations from machine 2 had sample variance $S_2^2 = 1.23$. Construct 90% confidence interval for the ratio $\frac{\sigma_2^2}{\sigma_1^2}$.

$$S_1^2 = 1.63 \quad S_2^2 = 1.23$$

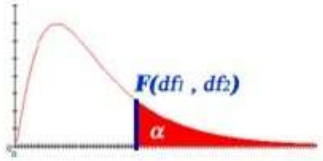
$$n_1 = 25 \quad n_2 = 16$$

$$F_{95\%,24,15} = 0.47 \quad F_{5\%,24,15} = 2.29$$

$$\frac{S_2^2}{S_1^2} = 0.75$$

$$[0.75 \times 0.47, 0.75 \times 2.29] = [0.3525, 1.7175]$$

F Table



F Table for α = 0.05

/	df_1=1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
df_2=2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.37
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

$$F_{\beta,n,m} = \frac{1}{F_{1-\beta,m,n}}$$

Testing one population variance

Consider following hypotheses,

$$H_0 : \sigma^2 = k \text{ , } H_1 : \sigma^2 \neq k$$

$$H_0 : \sigma^2 = k \text{ , } H_1 : \sigma^2 > k$$

$$H_0 : \sigma^2 = k \text{ , } H_1 : \sigma^2 < k$$

Where σ^2represents the population variance and $k > 0$. Then the test statistic is,

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

Testing one population variance

Ex:-

A manufacturer of hard safety hats for construction workers is concerned about the mean and the variation of the forces its helmets transmits to wearers when subjected to an external force. The manufacturer has designed the helmets so that the mean force transmitted by the helmets to the workers is μ pounds (or less) with a standard deviation to be less than 40 pounds. Tests were run on a random sample of 40 helmets and the sample mean and sample standard deviation were found to be 825 pounds and 48.5 pounds respectively. Do the data provide sufficient evidence at 5% level of significance to conclude that the population standard deviation exceeds 40 pounds.

$$H_0 : \sigma^2 = 40^2 = 1600 \text{ , } H_1 : \sigma^2 > 40^2 = 1600$$

Under H_0 ,

$$\chi^2_{cal} = \frac{39 \times 48.5^2}{1600} = 57.33$$

Testing one population variance

Ex:-

$$\chi^2_{0.05,39} = 54.57223$$

Since this is a one tail right tail test, since $\chi^2_{cal} > \chi^2_{0.05,39}$, H_0 will be rejected at 5% level of significance.

Conclusion – There is an evidence to say that the population standard deviation exceeds 40 pounds at 5% level of significance.

Comparison of Two variances

Consider following hypotheses,

$$H_0 : \sigma_x^2 = \sigma_y^2, H_1 : \sigma_x^2 \neq \sigma_y^2$$

$$H_0 : \sigma_x^2 = \sigma_y^2, H_1 : \sigma_x^2 > \sigma_y^2$$

$$H_0 : \sigma_x^2 = \sigma_y^2, H_1 : \sigma_x^2 < \sigma_y^2$$

$\sigma_x^2 = \sigma_y^2$ is equivalent for $\frac{\sigma_x^2}{\sigma_y^2} = 1$.

Where σ_x^2, σ_y^2 represent the population variance for the first population and for the second population respectively. Then the test statistic is,

$$F = \frac{S_x^2}{S_y^2}$$

Compare this F statistic value with the F table for degrees of freedoms (n-1) and (m-1) where n and m are the sample sizes. Rejection criteria is similar to the previous sections.

Comparison of Two variances

Ex:-

The variability in the amount of impurities present in a batch of chemicals used in a particular process depends on the length of time that the process is in operation. Suppose a sample of size 25 is drawn from the normal process which is to be compared to a sample of a new process that has been developed to reduce the variability of impurities.

	Sample 01	Sample 02
n	25	25
S^2	1.04	0.51

Consider the following hypothesis,

$$H_0 : \sigma_1^2 = \sigma_2^2, H_1 : \sigma_1^2 > \sigma_2^2$$

Where σ_1^2, σ_2^2 are variabilities of the old process and the new process respectively.

Comparison of Two variances

Ex:-

$$F_{cal} = \frac{S_1^2}{S_2^2} = \frac{1.04}{0.51} = 2.04$$

$$F_{5\%,24,24} = 1.98$$

Since this is a one tail right tail test as well as since $F_{cal} > F_{5\%,24,24}$, H_0 will be rejected at 5% level of significance.

Conclusion – There is an evidence to say that the variability in the new process is less than the variability in the original process at 5% level of significance.